Technical Report: MR Computing Network Infrastructure

1. Executive Summary

This technical report outlines the advanced cloud cluster topology, computing infrastructure, and network specifications designed to support efficient Magnetic Resonance (MR) image signal reconstruction and bleeding-edge imaging research. The architecture integrates on-premise Edge Nodes with massive, highly parallelized Cloud HPC instances and national supercomputing installations (Compute Canada) over specialized optical networks.

2. Network Topology and Computing Infrastructure

Architecture for efficient MR image signal reconstruction and imaging:

- Edge Nodes: Localized scanner gateways dedicated to immediate, real-time Fourier Transform approximations to offload primary scanner workloads.

- Compute Canada Integration: Seamless bursting capabilities to national Compute Canada HPC clusters (e.g., Niagara, Cedar GPU, Beluga GPU). These clusters handle heavy non-Cartesian gridding, complex iterative reconstructions, and deep learning AI inference logic.

- Network Path: Multi-tiered routing characterized by high-bandwidth and low-latency direct connections between DICOM nodes, local Edge clusters, and external research HPCs.

3. MR Research Computing & Cloud Topology

Advanced computing framework specifications targeted at pure MRI research:

- Hybrid Core: Secure on-premises edge nodes handling acute clinical data interoperating rapidly with multi-cloud resources without breaching PHI compliance.

- AWS/GCP Extensibility: Orchestrated hyper-scale Spot instance fleets configured specifically for massive parallelized parameter grid searches, radiomics feature extraction, and large-cohort foundation model AI training.

- Quantum Accelerators: Integration of experimental NVIDIA Quantum nodes for the bleeding-edge simulation of macroscopic spin-dynamics and quantum-assisted pulse sequences.

4. Detailed Network Specifications

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To facilitate the optimal transfer of enormous 4D MR datasets and raw k-space arrays, the network layer specifies:

- Protocol: TCP/IP optimized meticulously for jumbo frames (MTU 9000). Extensive use of RDMA over Converged Ethernet (RoCE) for cluster communications.
- Bandwidth: Persistent 100 Gbps dedicated optical links natively routed through the CANARIE network (and regional optical pipelines).
- Latency: Under 5ms RTT to regional foundational Compute Canada nodes to mimic local disk behavior over SANs.
- Security: Complete end-to-end AES-256-GCM point-to-point encryption, augmented with quantum-safe key exchanges across all boundaries.
- Reliability: 99.999% guaranteed uptime mediated by automated dual-redundant backbone paths routing over BGP.

5. Core Computing Infrastructure Hardware Specs

- Storage: Massively distributed Ceph File System utilizing a 10 PB high-IOPS NVMe tier specifically designated for active/hot clinical datasets.
- Compute Nodes: Fleet-scale compute-optimized instances sporting 128 vCPUs and at least 1TB RAM to digest traditional legacy workflows (FSL, FreeSurfer).
- GPU Acceleration: Dedicated enclosures of A100 and H100 instances exclusively serving dense multi-dimensional tensor operations and AI neural network reconstruction.
- Network Fabric: InfoBand 400Gbps switches guaranteeing unbounded internal cluster inter-node topology communications.